

Course: B. Tech.

Branch: Civil Engineering

Semester: V

Subject Code & Name: BTCVC501 Design of steel structure

Max Marks: 60

Date: 07/08/2023

Duration: 03.00 Hrs.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.
5. Use of IS:800-1984 and 2007, Steel table and IS:875-1987 is allowed

(Level/CO) Marks

Q.1 Solve Any Two of the following.

- A) Differentiate between the welded connection and riveted connection (CO1) 6
- B) Calculate the efficiency of single riveted double cover butt joint used to connect two plates of 12mm thick with 8mm cover plates. Using 20mm diameter power driven rivet. Assume pitch=60mm (CO2) 6
- C) The member in a roof truss consist of 2ISA80×50×8mm is connected to the 10mm thick gusset plate design a welded connection on three side of welds carries load of 460kN. (CO2) 6

Q.2 Solve Any one of the following (A or B).

A)

- 1) 2ISA 90×60×6mm connected to gusset plate of 12mm thick with 20mm diameter bolt. Determine the tensile strength of tension member connected 2ISA connected on either side of gusset plate with tacking rivet is provided (CO3) 6
- 2) A strut in a roof truss 2.5m long consist of 2ISA80×50×10mm discontinuous connected to 12mm thick gusset plate on same side of it. It is connected by using 2 bolts of 16mm diameter. Calculate the compressive strength of strut. (CO3) 6

OR

- B) Design suitable section for a beam having compression flange embedded into the concrete of clear span of 4m on 230mm thickness masonry wall. A beam carries dead load of 15kN/m due to floor above it and live load of 5kN/m assume self weight of beam is 1kN/m. (CO3) 12

Q.3 Solve Any Two of the following.

- A) Calculate the dead load per panel point for simple howe truss using (CO3) 6
following data:
Span of truss = 15m
Rise of truss= 3m
Intensity of covering= 120N/m^2
Intensity of purlin= 100N/m^2
Spacing of truss=4m
- B) Calculate the live load per panel point for simple howe truss (use the (CO1) 6
required data given in Q 3A)
- C) Explain and state the condition where the slab base, moment resisting base (CO1) 6
and grillage foundation is to be provided. Also draw the neat sketch.

- Q.4** Design the built up column consists of two channels placed back to back 6m (CO3) 12
to carry axial load of 1000KN. The column is effectively held in position at
both the ends but restrain against the rotation at one end. Also design the
single lacing system with bolted connection. $F_y=250\text{N/mm}^2$ (σ_{ac}) assumed=
 100 N/mm^2 , Angle of inclination of lacing bar=45 degree, Diameter of
bolt=20mm

Q.5 Solve Any Two of the following.

- A) Differentiate between elastic design and plastic design. (CO4) 6
- B) Define the following terms. (CO4) 6
1) Characteristic strength.
2) Characteristic load.
3) Shape factor.
- C) Define the term mechanism and explain the types of mechanism. (CO4) 6

*** End ***